PROJECT REPORT

Project Title: NeuroCanopy — AI-Driven Educational Ecosystem

Version: 1.0 (MVP)

Status: In Development

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1. EXECUTIVE SUMMARY

NeuroCanopy is an AI-powered educational ecosystem engineered to bridge the persistent gap between a student's academic potential and their actual performance. The platform goes beyond conventional study-aid tools — which merely organize tasks — by functioning as a "Digital Guardian" that actively enforces discipline through device-level control while simultaneously optimizing learning retention via bio-mimetic visualization, termed the "Knowledge Tree."

The system integrates two primary interfaces:

1. **A strict Android companion application** that manages distractions at the operating-system level.
2. **An intelligent React-based web dashboard** that provides analytics, adaptive scheduling, AI-powered voice assessments, and the signature Knowledge Tree visualization.

Together, these components form a closed-loop productivity system that enforces focus, adapts dynamically to academic schedules, and verifies concept mastery through active recall mechanisms.

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2. INTRODUCTION & BACKGROUND

The modern educational landscape is characterized by an unprecedented abundance of digital distractions. Students — particularly those in higher education — face constant temptation from social media, streaming services, and communication platforms. Traditional academic tools such as calendars, to-do lists, and note-taking apps are inherently passive; they require the student to possess self-discipline to use them effectively.

NeuroCanopy was conceived from the observation that the three primary barriers to academic success in the digital age are:

1. Digital fragmentation and inability to sustain focus.
2. Rigid, unadaptive scheduling that crumbles under real-world pressure.
3. Passive consumption of learning material without active verification of understanding.

The project aims to address each barrier with a purpose-built subsystem, unified under a single ecosystem that synchronizes in real time between mobile and web interfaces.

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3. PROBLEM STATEMENT

Current educational productivity tools fail to address the three primary causes of academic underperformance in the digital age:

3.1 Digital Fragmentation

Students lose an average of 3–4 hours daily to social media distractions and context switching between productive and non-productive applications. Existing solutions rely on voluntary self-control, which is insufficient for sustained focus.

3.2 Static Planning

Traditional calendars, planners, and scheduling tools do not adapt to real-time disruptions such as burnout, sudden exam announcements, or unforeseen academic obligations. This leads to "schedule collapse," where the student abandons the plan entirely and enters a state of panic-driven, unstructured cramming.

3.3 Passive Learning

Students frequently engage in passive reading without retaining information. There is a critical absence of on-demand active recall mechanisms — such as oral viva examinations — available outside the classroom to verify concept mastery and reinforce memory.

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4. OBJECTIVES

The primary objectives of the NeuroCanopy project are:

- **O1.** Develop an Android companion application capable of enforcing distraction-free study sessions through OS-level app-blocking and device administration privileges.
- **O2.** Build an intelligent, React-based web dashboard that provides:
  - Real-time academic analytics (mastery index, effort logs, etc.)
  - A bio-mimetic "Knowledge Tree" visualization reflecting syllabus progress and memory retention.
  - An adaptive AI-driven scheduling engine.
  - A comprehensive syllabus management interface.
- **O3.** Integrate large language models (Gemini) for:
  - Automated syllabus parsing and topic extraction.
  - AI-conducted voice viva examinations with speech analysis.
  - Intelligent doubt resolution and summary generation.
- **O4.** Implement a gamification layer (Streak System, Redemption Store) that balances strict enforcement with positive reinforcement.
- **O5.** Ensure real-time synchronization between mobile and web interfaces through Firebase Firestore.

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5. SOLUTION ARCHITECTURE

NeuroCanopy deploys a Closed-Loop Productivity System comprising three interlocking subsystems:

```
ENFORCEMENT (Mobile App)
────────────────────────
Removes the option of distraction by locking the digital
environment during study hours. Utilizes Device Admin,
Draw-Over, and Usage Access permissions.

ADAPTABILITY (AI Dashboard)
───────────────────────────
Ingests schedules from informal channels and restructures
study plans dynamically based on "Panic," "Alert," or
"Smooth" preparation states.

VERIFICATION (AI Assessment Engine)
───────────────────────────────────
Replaces passive reading with active "Voice Vivas" — AI-
conducted oral examinations that grade verbal answers for
accuracy, clarity, and confidence.
```

Data flows bidirectionally between all three subsystems via Firebase Firestore, ensuring that enforcement decisions on the mobile app are informed by analytics from the dashboard, and vice versa.

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6. PROJECT SCOPE

6.1 In-Scope

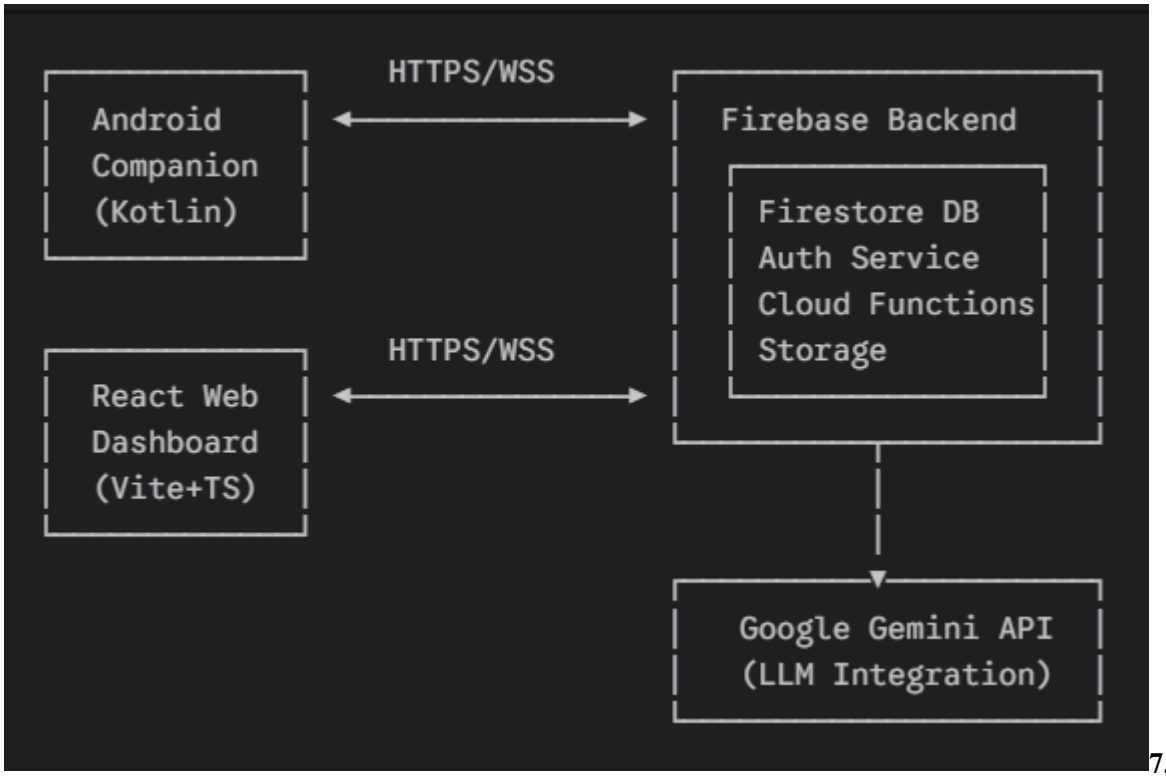
- **Android Admin Application:** A non-uninstallable companion app with "Draw Over" and "Usage Access" permissions for enforcement of focus sessions.
- **Web-Based Command Center:** A React-based dashboard for analytics, the Knowledge Tree visualization, Voice Viva assessments, and the Redemption Store.
- **AI Backend:** Integration of LLMs (Google Gemini API) for syllabus parsing, viva question generation, speech-to-text analysis, and intelligent scheduling.
- **Real-Time Database:** Firebase Firestore implementation for instant data synchronization between the mobile blocker and web dashboard.

6.2 Out-of-Scope (Phase 1)

- **iOS Application:** Due to OS-level restriction limitations, Apple's sandboxing policies prevent the device-administration features required by the enforcement module.
- **Physical Hardware Integration:** Smart wearables and IoT-based monitoring are deferred to future development phases.

7. SYSTEM DESIGN & TECHNICAL ARCHITECTURE

7.1 High-Level Architecture



7.2 Technology Stack

Layer	Technology	Purpose
Frontend (Web)	React 18 + TypeScript	Main web dashboard UI
Build Tool	Vite 7.x	Fast development builds
Styling	Tailwind CSS 3.4	Utility-first CSS framework
UI Components	Radix UI + shadcn/ui	Accessible component library
State Management	TanStack React Query	Async state & caching
Visualization	D3.js v7	Knowledge Tree rendering
Animations	Framer Motion	Page & element transitions

Layer	Technology	Purpose
Charts	Recharts	Analytics charts & graphs
Forms	React Hook Form + Zod	Form handling & validation
Routing	React Router DOM v6	Client-side routing
Backend (BaaS)	Firebase v12	Authentication, Firestore, Cloud Functions, Storage
AI Integration	Google Gemini API	Text parsing, speech analysis, question generation
PDF Processing	PDF.js	Syllabus & note extraction
Mobile (Android)	Kotlin / React Native	Distraction enforcement app
Security	Firebase Auth + RLS	Row-level security & auth

7.3 Directory & Module Structure

```
V6/
├── documents/          # PRD, feature docs, project report
├── public/             # Static assets (icons, manifest)
├── src/
│   ├── components/
│   │   ├── auth/       # Authentication & registration flows
│   │   │   └── steps/  # Multi-step registration wizard
│   │   ├── dashboard/  # Dashboard widgets & analytics cards
│   │   ├── landing/    # Landing page sections (Hero, etc.)
│   │   └── ui/         # Reusable UI primitives (49 components)
│   ├── contexts/       # React context providers
│   ├── hooks/          # Custom React hooks (8 hooks)
│   ├── lib/            # Utility libraries (Firebase init, etc.)
```

```
│   ├── pages/              # Route-level page components (14 pages)
│   │   ├── Index.tsx        # Landing page
│   │   ├── Login.tsx        # User authentication
│   │   ├── Register.tsx     # Multi-step registration
│   │   ├── Dashboard.tsx    # Main analytics dashboard
│   │   ├── KnowledgeTreePage # Bio-mimetic tree visualization
│   │   ├── Syllabus.tsx     # Syllabus management & upload
│   │   ├── VoiceViva.tsx    # AI-powered voice assessments
│   │   ├── Schedule.tsx     # Adaptive scheduling calendar
│   │   ├── Chat.tsx         # AI learning assistant
│   │   ├── Profile.tsx      # User profile management
│   │   ├── Settings.tsx     # Application settings
│   │   └── ...              # Password reset, error pages, etc.
│   └── utils/               # Shared utility functions
├── index.html               # Application entry point
├── vite.config.ts           # Vite build configuration
├── tailwind.config.ts       # Tailwind CSS configuration
├── tsconfig.json            # TypeScript compiler options
├── package.json             # Dependencies & scripts
└── storage.rules            # Firebase Storage security rules
```

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8. FEATURE SPECIFICATIONS

8.1 Module I — Mobile Companion (Android)

Designed for high-accountability enforcement and data ingestion.

- **8.1.1 Intelligent Schedule Parsing:** * Utilizes notification access to monitor communication platforms (e.g., WhatsApp groups) for keywords related to exams, deadlines, or schedule changes.
    - **Action:** Automatically prompts the user to add detected dates to the central database or upload relevant documents immediately.
- **8.1.2 Distraction Management Protocol:**
    - Enforces a strict "Focus Mode" during scheduled study sessions or when daily targets are unfulfilled.

    - **Blocking Logic:** Temporarily restricts access to social media and entertainment applications. Includes a configurable "buffer" allowance (5–10 minute emergency windows) to maintain safety without compromising focus.

- **8.1.3 Persistent Accountability Alerts:**
    - Utilizes "Draw Over Other Apps" permissions to display intrusive reminders when critical tasks are neglected (e.g., failure to upload a syllabus).
    - **Escalation:** Alerts become persistent and non-dismissible upon repeated user non-compliance.
- **8.1.4 Device Administration Security:**
    - Operates as a Device Admin app to prevent unauthorized uninstallation during active academic terms, ensuring consistent adherence to the study regimen.

8.2 Module II — Intelligent Web Platform (AI Dashboard)

The central command center for analytics, visualization, and learning management. Implemented in React + TypeScript with Vite.

- **8.2.1 Dynamic Pacing Engine:** Categorizes user preparation status in real time:
    - **Smooth:** User is on track with the syllabus.
    - **Alert:** Mild backlog detected; schedule compresses.
    - **Panic:** Critical time deficit; AI auto-prioritizes high-yield topics and ignores low-weightage material to optimize remaining time.
- **8.2.2 Comprehensive Analytics Suite:**
    - **Concept Mastery Index:** % of concepts mastered vs. total.
    - **Effort Estimation:** Meaningful study hours (derived from Focus Mode duration and topic completion).
    - **Viva Performance Metrics:** Average viva scores, session frequency, and per-topic breakdown.
    - **Review Radar:** Identifies "At-Risk" topics requiring immediate review based on spaced-repetition decay.
    - **Subject Proficiency Graph:** Radar chart visualizing strengths and weaknesses across all subjects.
- **8.2.3 Visualized Knowledge Graph (The Tree):**
    - **Organic Progress Visualization:** The syllabus is rendered as a living D3.js-powered tree.
        - *Growth:* New topics appear as buds/small leaves.
        - *Mastery:* Completed topics become vibrant green leaves.
        - *Decay:* Unrevised topics turn brown and eventually "fall," visually signaling memory decay (spaced repetition enforcement).

    - **Scale:** The tree's complexity and size dynamically adjust to match the total number of topics in the curriculum.

- **8.2.4 Adaptive Scheduler (Quantum Calendar):**
    - A dynamic calendar that updates in real time.
    - If a study block is missed or a new exam is announced, the AI automatically reshuffles remaining tasks to ensure syllabus coverage without manual replanning.
- **8.2.5 Syllabus Commander & Resource Vault:**
    - **Curriculum Tracking:** Displays total courses, progress grades, and topic-level breakdown.
    - **Resource Library:** Users upload learning materials (PDFs, PPTs, handwritten notes). The AI parses these to auto-populate the topic list.

8.3 Module III — AI Assessment & Tutor Engine

The interactive layer for testing and concept reinforcement.

- **8.3.1 Secure Assessment Environment (Voice Viva & Quiz):**
    - **Anti-Cheat Integrity:** Enforces a locked browser/window state during all assessments.
        - *Violation Logic:* Copy-pasting, tab switching, or window minimizing triggers immediate session cancellation.
        - *Punitive Measures:* Repeated violations trigger a system-wide lock on leisure apps via the mobile companion app.
    - **AI Viva Examiner:** Conducts oral examinations on scheduled topics.
        - *Speech Analysis:* Evaluates answer accuracy, speech clarity, confidence level, and pronunciation quality.
- **8.3.2 Interactive Learning Assistant:**
    - **Doubt Resolution:** Provides an interface for topic-specific clarification questions before or after viva.
    - **Summary Generation:** Creates concise summaries of uploaded PDFs for quick revision during Panic Mode.
- **8.3.3 Gamification & Redemption System:**
    - **Streak System:** Tracks consecutive days of meeting study goals.
    - **Redemption Store:** Users "purchase" lost privileges (social media time) or restore "fallen leaves" by completing extra credit quizzes — balancing the strict enforcement model with positive reinforcement.

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9. DATABASE DESIGN

NeuroCanopy utilizes Firebase Firestore as its primary database. The database is organized into the following principal collections:

Collection	Description
users/	User profile data, preferences, registration info, timetable, and academic configuration.
users/{uid}/courses/	Enrolled courses with associated knowledge trees.
users/{uid}/vivaResults/	Historical viva assessment scores & transcripts.
users/{uid}/streaks/	Gamification data (daily streak counters).
users/{uid}/focusSessions/	Logged Focus Mode sessions with duration metrics.
syllabus/	Parsed syllabus data with topic hierarchy, weights, and mastery states used by the Knowledge Tree.
schedules/	AI-generated and user-modified study schedules with real-time pacing state (Smooth/Alert/Panic).
resources/	User-uploaded learning materials (PDFs, notes) with metadata and AI-extracted summaries.
chatMessages/	Conversation history with the AI Learning Assistant.
redemptionStore/	Available rewards and user transaction history.

All collections enforce strict Row-Level Security (RLS) rules to ensure that users can only read and write their own data.

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10. SECURITY & PRIVACY CONSIDERATIONS

10.1 Authentication

- Firebase Authentication with email/password and Google Sign-In (OAuth 2.0).
- Password reset flows via secure email links.

10.2 Data Privacy

- All user data is isolated via Firestore security rules.
- API keys are managed via environment variables (.env) and are never committed to version control (.gitignore enforced).
- Strict Row-Level Security policies on Firestore ensure users cannot access other users' data.

10.3 Assessment Integrity

- The Voice Viva module enforces a locked assessment environment with tab-switch and copy-paste detection.
- Violations are logged and trigger escalating consequences.

10.4 Device Security

- The Android companion app operates under Device Admin privileges to prevent unauthorized removal during active academic periods.

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11. KEY PERFORMANCE INDICATORS (KPIs)

The success of NeuroCanopy will be measured by the following metrics:

KPI #	Metric	Description
KPI-1	Retention Rate	Percentage of "Green Leaves" maintained on the Knowledge Tree on a week-over-week basis.
KPI-2	Distraction Reduction	Measurable decrease in social media usage statistics during active Focus Mode blocks.
KPI-3	Viva Proficiency	Improvement in average Voice Viva scores over a rolling 30-day period.
KPI-4	Schedule Adherence	Percentage of planned study blocks completed vs. scheduled (target: >80% in Smooth state).
KPI-5	User Engagement	Daily active usage of the dashboard and average session duration.

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12. DEVELOPMENT METHODOLOGY & TIMELINE

12.1 Methodology

- Agile/Iterative development with weekly sprints.
- Version-controlled via Git with modular branching strategy.
- Continuous integration through Vite build system.

12.2 Development Phases

Phase	Deliverable	Status
P1	Project Setup, Authentication, Registration	Completed
P2	Dashboard & Analytics Suite	Completed
P3	Knowledge Tree (D3.js Visualization)	Completed
P4	Syllabus Commander & Resource Vault	Completed
P5	Voice Viva & AI Assessment Engine	Completed
P6	Adaptive Scheduler (Quantum Calendar)	Completed
P7	AI Chat / Learning Assistant	Completed
P8	Gamification & Redemption Store	In Progress
P9	Android Companion App	Planned
P10	Integration Testing & Launch	Planned

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13. CHALLENGES & MITIGATION STRATEGIES

Challenge	Mitigation Strategy
AI response latency for vivas	Implement streaming responses and loading state indicators; use Gemini Flash for lower-latency interactions.
Speech-to-text accuracy	Use Google's Gemini speech analysis and provide user feedback for unclear audio; allow re-recording.
Battery / performance impact of Android blocker	Optimize Focus Mode service with minimal CPU footprint; use Android WorkManager for background scheduling.
Knowledge Tree rendering at scale (100+ topics)	Implement lazy rendering and node clustering in D3.js; virtualize off-screen branches.

Challenge	Mitigation Strategy
Schedule collapse edge cases	Design fallback logic for extreme Panic Mode scenarios; guarantee a minimum viable study plan always exists.
User resistance to enforcement	Balance strict measures with the Redemption Store incentive system; allow customizable strictness levels.

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14. FUTURE ENHANCEMENTS (Phase 2+)

- **iOS Companion App:** Explore MDM-based enforcement or guided access APIs as Apple's platform evolves.
- **Smart Wearable Integration:** EAR-based attention tracking via smartwatch sensors for automatic focus detection.
- **Collaborative Study Rooms:** Real-time virtual study groups with shared Knowledge Trees and peer accountability.
- **Institutional Dashboards:** Aggregate analytics for educators and institutions to monitor cohort-level progress.
- **Advanced NLP Tutoring:** Context-aware tutoring that references the student's uploaded materials and past viva transcripts.
- **Offline Mode:** Full offline functionality for both web and mobile platforms with background sync on reconnection.
- **Multi-Language Support:** Localization for regional Indian languages to expand accessibility.

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15. CONCLUSION

NeuroCanopy represents a paradigm shift in educational technology — moving from passive organizational tools to an active, AI-powered enforcement and learning ecosystem. By combining device-level distraction management with intelligent adaptive scheduling and rigorous voice-based assessment, the platform addresses the three core deficiencies of existing academic tools: fragmented focus, rigid planning, and passive learning.

The MVP, currently in active development, has successfully delivered its web-based command center with functional authentication, dashboard analytics, Knowledge Tree visualization, syllabus management, voice viva assessments, adaptive scheduling, and an AI chat assistant. The remaining work focuses on gamification refinements and the development of the Android companion application.

Upon completion, NeuroCanopy aims to demonstrate measurable improvements in student focus, study consistency, and concept retention — validated through the defined KPIs of leaf retention rates, distraction reduction metrics, and viva proficiency scores.

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16. REFERENCES & APPENDICES

References:

1. Google Gemini API Documentation — ai.google.dev
2. Firebase Documentation — firebase.google.com/docs
3. React.js Official Documentation — react.dev
4. D3.js Documentation — d3js.org
5. Vite Build Tool — vitejs.dev
6. Tailwind CSS — tailwindcss.com
7. Radix UI / shadcn/ui — ui.shadcn.com

Appendices:

- **A.** Product Requirements Document (PRD.txt)
- **B.** Feature Specification Document (feature section.txt)
- **C.** Registration Flow Specification (Registration section.txt)